

SIMATS ENGINEERING ASSESSMENT
TOOL 3 – VIVA QUESTIONS

Course: Ethical Hacking	Code: ITA1408	CO: C05
Duration: 1	Total Marks: 100	Reg No / Name : Ranjithkumar 192411119

COURSE OUTCOME COVERED

C05: Analyze vulnerabilities in Windows and Linux systems and evaluate security weaknesses in messaging, web, and mobile applications using appropriate exploitation techniques and hacking tools. (BT5)

Code of Conduct:

I Ranjithkumar (912411119) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prky

Instructions: Answer all questions. Each question carries 4 marks.

1. How would you evaluate the security posture of a Windows system after identifying multiple open services?
 2. Which factors would you consider when choosing a hacking tool for Windows vulnerability assessment, and why?
 3. How would you evaluate the security risk associated with an exposed RPC service?
 4. Why are Windows Null Sessions considered a security weakness, and how would you assess their impact?
 5. A Windows shared folder allows unauthorized users to access sensitive files. How would you evaluate the severity of this vulnerability?
 6. How would you distinguish between a vulnerable Windows service and a misconfigured Windows service during security testing?
 7. How would you evaluate the effectiveness of information-gathering techniques against a Windows target?
 8. If a Windows vulnerability can be exploited but requires local access, how would you classify its security impact compared with a remotely exploitable vulnerability?
 9. How would you evaluate the security risks of unconnected or unnecessary services running on a Linux system?
 10. Why are rhosts and hosts.equiv considered security risks in Linux? How would you evaluate their impact on system security?
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11. How would you assess whether an NFS configuration exposes sensitive Linux files to unauthorized users?
 12. A Linux file has incorrect permissions that allow unauthorized modification. How would you evaluate the resulting security risk?
 13. How would you compare the security impact of weak file permissions with an unpatched Linux vulnerability?

14. How would you evaluate whether a buffer overflow vulnerability in a Linux application is practically exploitable?
15. Which factors would you consider before selecting an exploitation technique for a Linux vulnerability?
16. How would you evaluate the security of a messaging application that stores user credentials or messages insecurely?
17. A web application is vulnerable to SQL injection. How would you evaluate the severity of this vulnerability and its possible consequences?
18. How would you determine whether a web application vulnerability represents a critical security risk or a low-risk finding?
19. How would you evaluate the security of a mobile application using a tool such as MobSF?
20. A mobile application requests unnecessary permissions such as contacts, microphone, and location. How would you evaluate the associated security risk?
21. How would you evaluate the risk of hardcoded API keys or passwords discovered during mobile application analysis?
22. If the same vulnerability is found in both a web application and a mobile application, how would you compare their security impact?
23. How would you select appropriate hacking tools for assessing Windows, Linux, web, and mobile applications?
24. After performing vulnerability exploitation, how would you evaluate whether the exploitation technique actually demonstrates a significant security weakness?
25. You have identified vulnerabilities in Windows, Linux, messaging, web, and mobile applications. How would you prioritize them and prepare a professional security report?

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Mark
Understanding of Problem Context	20	Clearly understands the security problem and accurately identifies vulnerabilities, risks, and requirements	Good understanding of the security problem with minor gaps	Basic understanding; some security requirements or vulnerabilities are unclear	Poor understanding or misinterpretation of the security problem	
Application of Concepts	30	Accurately applies appropriate ethical hacking concepts, tools, and exploitation techniques to analyze complex security problems	Applies relevant security concepts and tools correctly with minor errors	Applies basic concepts with limited accuracy or tool selection	Fails to apply appropriate security concepts or tools	
Problem-Solving Approach	20	Demonstrates a logical, well-structured, systematic, and efficient approach to identifying and evaluating security weaknesses	Mostly logical and systematic approach with minor gaps	Basic approach with some inconsistencies in vulnerability analysis	Illogical, unclear, or inappropriate approach to solving the security problem	

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Accuracy of Solution	20	Provides a completely correct security analysis with appropriate justification, evidence, and suitable prevention	Mostly correct analysis with minor errors or gaps	Partially correct analysis with limited justification	Incorrect or incomplete security analysis	
Clarity	10	Provides clear, well-organized, technically sound, and properly explained responses using appropriate security	Understandable response with minor clarity or terminology issues	Limited clarity and explanation; uses technical terms inconsistently	Poorly presented, unclear, and difficult to understand	
Total	100					/100

- 1) Check open services, missing patches, weak configurations, user permissions, and vulnerabilities.
- 2) Accuracy, reliability, compatibility, safety, and ease of use.
- 3) An exposed Rpc service can allow attacks to exploit vulnerabilities or gain unauthorized access.
- 4) They may allow unauthorized users to obtain system or network information.
- 5) It is serious if sensitive files can be read, modified, or deleted without authorization.
- 6) A vulnerable service has a software weakness; a misconfigured service is insecure because of incorrect settings.
- 7) It helps identify open ports, services, versions, users, and possible vulnerabilities.
- 8) A local vulnerability requires access to the system, while a remote vulnerability can be exploited over a network.
- 9) They increase the attack surface and may provide unwanted entry points.
- 10) Incorrect configurations can allow unauthorized systems or users to access Linux resources.
- 11) Check exposed directories, allowed hosts, permissions, and authentication settings.

- 12) unauthorized users may read, modify, or delete important files
- 13) Both are risks, but an unpatched memory exploitable vulnerability can be more critical
- 14) Check whether the vulnerable code is reachable and whether security protections prevent exploitation
- 15) Vulnerability type, software version, system protections, access level, and environment.
- 16) It can expose passwords or messages and lead to account compromise
- 17) It can allow unauthorized database access, data theft, modification, or deletion
- 18) Consider exploitability, affected data, user impact, and possible business damage
- 19) Check permissions, code, storage, network security, APIs, and configuration
- 20) They may expose sensitive user data such as contacts, microphone, or location.

- 1) Attackers can extract and misuse them to access service or data.
- 22) Compile affected users, exposed data, exploitability and possible damage.
- 23) Choose tools based on the platform, vulnerability type, accuracy, and testing requirements.
- 24) Confirm whether the vulnerability actually allowed unauthorized access or impact without causing unnecessary damage.
- 25) Rank them by severity and impact, then report the vulnerability, evidence, risk, and recommended fix.